



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

YEARLY EXAMINATION
PRELIMINARY ASSESSMENT TASK 4

Year 11

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2½ hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Show all necessary working in Questions 11 – 16
- Write your Name **and** Class Teacher on each of the writing booklet(s) submitted
- **ASSESSMENT WEIGHTING:** 40%

Name:

Class Teacher:

*Do **NOT** write solutions on this question paper. Any working on the question paper will **NOT** be marked.*

Total marks – 100

Section I

Pages 3 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 6 – 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 15 minutes for this section

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Section I 10 marks

- Circle the correct response on the answer sheet provided
 - Each question is worth 1 mark
-

1 Use your calculator to evaluate $\frac{112.8}{16.1 \times 2.93}$ correct to 3 significant figures

- (A) 2.39 (B) 2.391 (C) 20.5 (D) 20.528

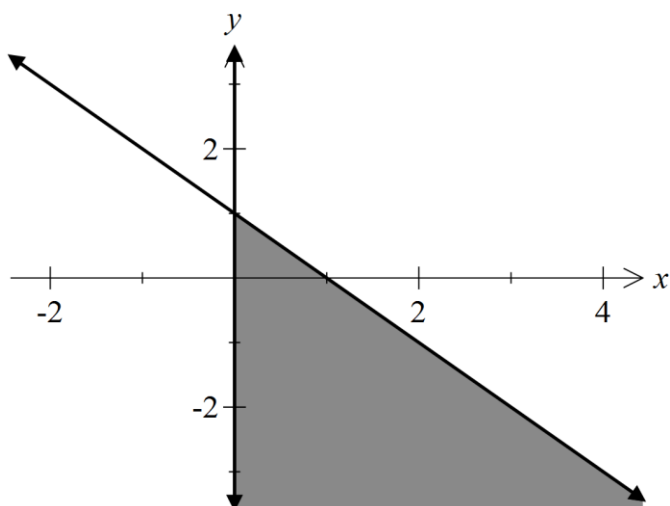
2 Simplify $\frac{a^6 \times (a^8)^2}{a^2}$

- (A) a^8 (B) a^{11} (C) a^{14} (D) a^{20}

3 Make y the subject of the equation $2y = 1 - xy$

- (A) $y = \frac{1-x}{3}$ (B) $y = \frac{1}{2+x}$ (C) $y = \frac{1-xy}{2}$ (D) $y = \frac{1-2y}{x}$

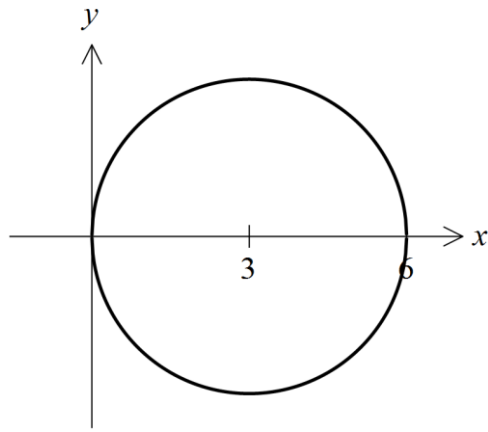
4



Which pair of inequalities best describes the shaded region in the above diagram

- (A) $x + y \leq 1$ and $x \geq 0$ (B) $x + y \leq 1$ and $y \geq 0$
(C) $x + y \geq 1$ and $x \geq 0$ (D) $x + y \geq 1$ and $y \geq 0$

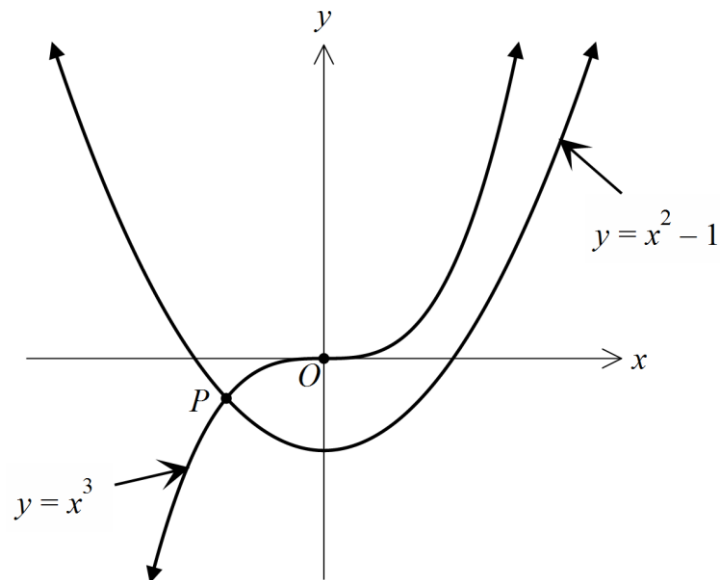
5



The equation of the circle drawn above is

- (A) $(x+3)^2 + y^2 = 9$
- (B) $(x+3)^2 + y^2 = 36$
- (C) $(x-3)^2 + y^2 = 9$
- (D) $(x-3)^2 + y^2 = 36$

6



The graphs of $y = x^3$ and $y = x^2 - 1$ intersect at P .

Which equation gives the x co-ordinate of P ?

- (A) $x^2 - x^3 + 1 = 0$
- (B) $x^3 - x^2 + 1 = 0$
- (C) $x^3 + x^2 - 1 = 0$
- (D) $x^3 + x^2 + 1 = 0$

7 Factorise $y^2 - (x+3)^2$

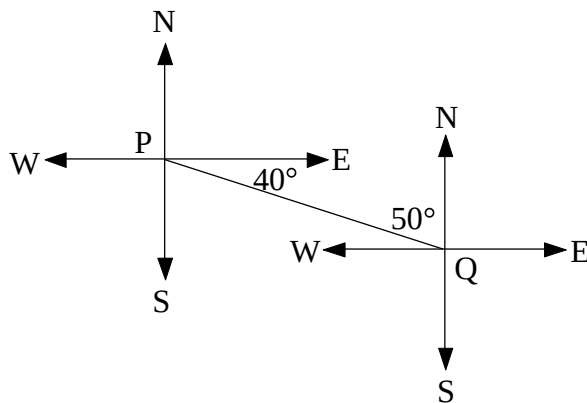
(A) $(y-x+3)(x+y+3)$

(B) $(y-x-3)(x+y+3)$

(C) $(y-x+3)^2$

(D) $(y-x-3)^2$

8



The bearing of P from Q is

(A) 040° (B) 050° (C) 130° (D) 310°

9 Given $180^\circ < \theta < 270^\circ$, which of the following is correct?

(A) $\sin \theta > 0$ and $\tan \theta > 0$

(B) $\sin \theta > 0$ and $\tan \theta < 0$

(C) $\sin \theta < 0$ and $\tan \theta > 0$

(D) $\sin \theta < 0$ and $\tan \theta < 0$

10 The expression $\frac{b + \sqrt{2b}}{\sqrt{b}} =$

(A) $\sqrt{2} + b$ (B) $\sqrt{3b}$ (C) $3\sqrt{b}$ (D) $\sqrt{2} + \sqrt{b}$

End of Section I

Section II 90 marks

- Begin each question in a new writing booklet.
 - Write your name **and** your class teacher's name on each answer booklet used
 - Show all necessary working
-

Question 11 (15 marks)

(a) Fully factorise $x^4 - 27x$. 2

(b) Solve $|3x - 4| < 8$. 2

(c) Graph the solution to $x^2 - 11x + 24 > 0$ on a real number line. 2

(d) Simplify $\frac{4}{m^2 - 4} - \frac{1}{m - 2}$. 3

(e) Solve simultaneously $\begin{cases} 3a - 4b = 11 \\ 2a + 3b = -4 \end{cases}$. 3

(f) Find the values of a and b such that $\frac{\sqrt{5}}{\sqrt{5} - 2} = a + \sqrt{b}$. 3

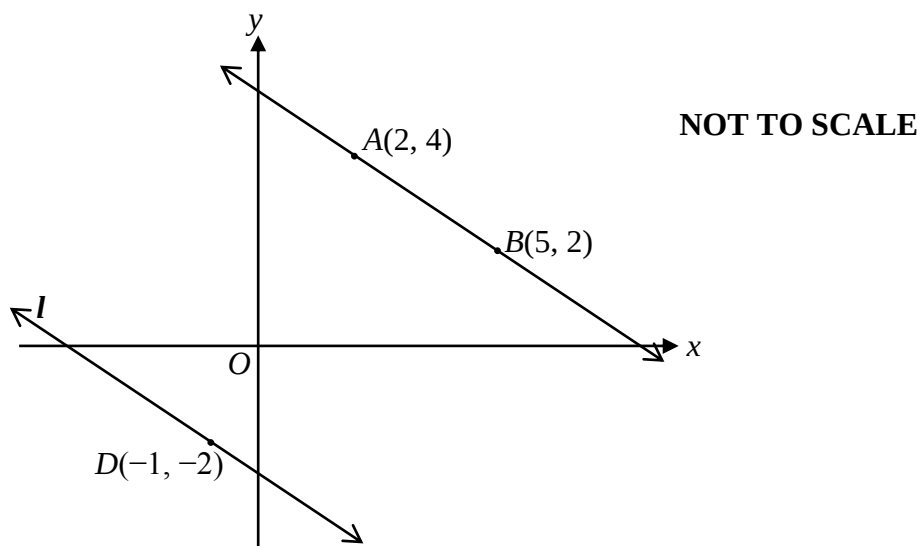
Question 12 (15 marks)**Begin a NEW answer booklet**

(a) (i) Sketch the graph of $y = \frac{1}{x-3}$, showing its main features. 2

(ii) State the domain for the function $y = \frac{1}{x-3}$. 1

(b) Given $f(x) = x^{-2} + \frac{2}{x}$, simplify $f\left(\frac{1}{a}\right)$. 2

(c) Consider the diagram below with points $A(2, 4)$ and $B(5, 2)$.



(i) Find the gradient of the line through AB . 2

(ii) Show that the equation of the line l parallel to the line AB and passing through $D(-1, -2)$ is $2x + 3y + 8 = 0$. 2

(iii) Show the distance OA is $2\sqrt{5}$ units. 2

(iv) Find the equation of the circle with centre at the origin and passing through A . 1

(v) The radius AO of the circle in part (iv) is produced to C on the circumference. Find the co-ordinates of the point C . 2

(vi) State the perpendicular distance from the point B to the line l . 1

Question 13 (15 marks)**Begin a NEW answer booklet**

(a) Solve the following equations in the domain $0^\circ \leq \theta \leq 360^\circ$.

(i) $\sin \theta = \frac{1}{2}$ 2

(ii) $\cos 2\theta = 0$ 2

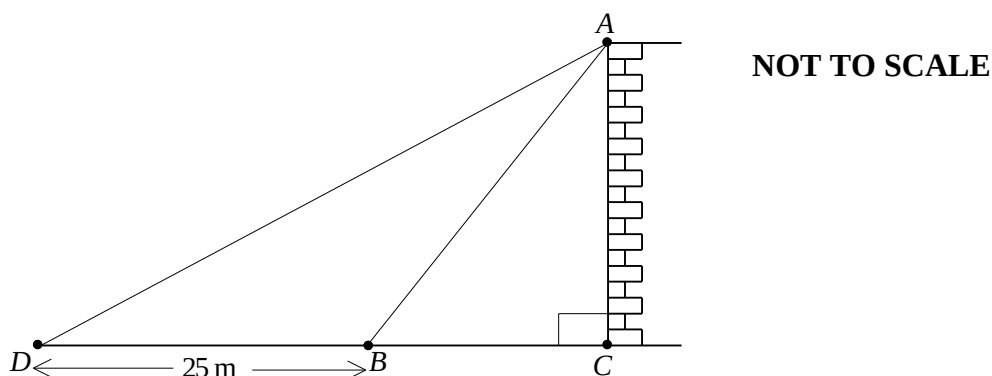
(iii) $\tan \theta = -\sqrt{3}$ and $\sin \theta < 0$ 2

(b) Simplify $\frac{\sin^2 \theta}{1 - \cos \theta} - \frac{\sin^2 \theta}{1 + \cos \theta}$ 2

(c) (i) Show that $f(x) = x\sqrt{4-x^2}$ is an odd function. 2

(ii) Hence, or otherwise, if $f(x-1) = \sqrt{3}$, state the value of $f(1-x)$. 1

(d)



A person at D observes the angle of elevation of the top of a building A to be 28° .

The person walks from D , 25m towards the building across a horizontal plane to a point B and finds the angle of elevation now to be 32° .

Copy the diagram into your writing booklet and label all the given information.

(i) Explain why $\angle DAB = 4^\circ$. 1

(ii) Using the Sine Rule, find correct to one decimal place, the length BA . 2

(iii) Find the height of the building, correct to one decimal place. 1

Question 14 (15 marks)**Begin a NEW answer booklet**

(a) Find real constants A , B and C such that $2x^2 + x + 1 \equiv Ax^2 + Bx + C$. 3

(b) The roots of the equation $2x^2 - 7x + 12 = 0$ are α and β . Find the value of:

(i) $\alpha + \beta$ 1

(ii) $\alpha\beta$ 1

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ 2

(iv) $\alpha^2\beta + \beta^2\alpha$ 2

(v) $\alpha^2 + \beta^2$ 2

(c)(i) Write down in simplest form, the discriminant
 $4k+1$ $x^2 - 2$ $k+1$ $x - (1-2k) = 0$. 2

Hence, find the value of k such that the roots of $k-1$ $x^2 + k+1$ $x - 2 = 0$ are equal.

(ii) Find the value of k in the equation $k-1$ $x^2 + k+1$ $x - 2 = 0$ so that one root is the reciprocal of the other. 2

(**Hint:** Let the roots be α and $\frac{1}{\alpha}$)

Question 15 (15 marks)**Begin a NEW answer booklet**

(a) A function is defined as $f(x) = \begin{cases} 1 - x & x < 0 \\ x^2 + 1 & x \geq 0 \end{cases}$

(i) Find the value of $f(-2) + f(3)$. 2

(ii) Sketch the graph of $y = f(x)$. 2

(b) If $y = 5x^4 + \frac{3}{x} + 2\sqrt{x}$, find $\frac{dy}{dx}$. 3

(c) Use the quotient rule to find $\frac{dy}{dx}$ if $y = \frac{x^2 + 1}{3 - x}$. 2

(d) If $f(x) = (2x - 5)^4$ find the value of x such that $f'(x) = 0$. 2

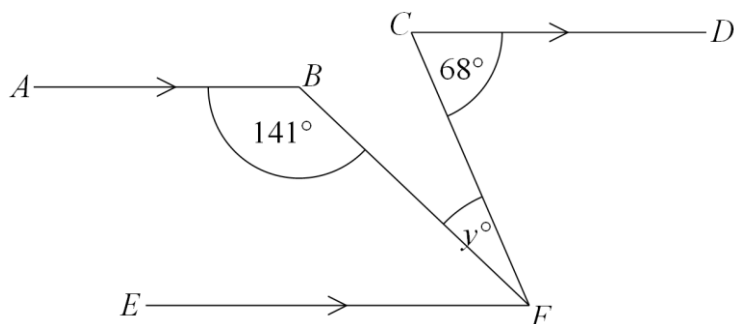
(e) Find the equation of the tangent to the curve $y = x^3 + 5x - 4$ at the point where $x = 1$. Leave your answer in the form $y = mx + b$. 2

(f) Find the value of $\lim_{x \rightarrow 3} \left(\frac{x^2 - 9}{x - 3} \right)$. 2

Question 16 (15 marks)

Begin a NEW answer booklet

(a)



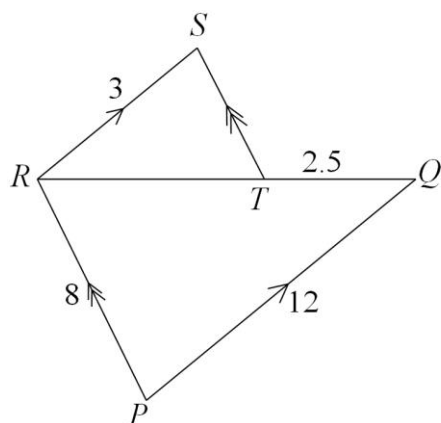
NOT TO SCALE

Copy this diagram into your writing booklet.

Using the information on the diagram above, find the value of y , giving reasons for your answer.

3

(b)



NOT TO SCALE

(all measurements are in cm)

Copy this diagram into your writing booklet.

In the diagram above $RS \parallel PQ$ and $ST \parallel RP$.

(i) Prove that $\triangle RST$ is similar to $\triangle QRP$.

3

(ii) Find the length of ST , giving a reason for your answer.

2

(c) Find the exact value of $\sin 150^\circ \sec(-150^\circ)$.

2

(Question 16 continues on the following page)

Question 16 continued

- (d) Sketch the graph of the parabola $x^2 - 6x + 9 + 12y = 0$, showing **and** stating the coordinates of its vertex, focus and the equation of the directrix. **3**
- (e) The point $P(x, y)$ is equidistant from the lines $x = 2$ and $y = -4$. Find the equation of the locus of P . **2**

End of Paper



TRINITY GRAMMAR SCHOOL

2012, Year 11 Mathematics

Yearly Examination, Preliminary Assessment Task 4

SECTION I

ANSWER SHEET

Name:

Class Teacher:

Be sure to write your answers for **Section I** on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. **Mark only one answer for each question.**

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D



TRINITY GRAMMAR SCHOOL

2012, Year 11 Mathematics

Yearly Examination, Preliminary Assessment Task 4

SECTION I

ANSWER SHEET

Name: *Solutions*

Class Teacher:

Be sure to write your answers for **Section I** on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. *Mark only one answer for each question.*

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | ☒ A | ☐ B | ☐ C | ☐ D |
| 2 | ☐ A | ☐ B | ☐ C | ☒ D |
| 3 | ☐ A | ☒ B | ☐ C | ☐ D |
| 4 | ☒ A | ☐ B | ☐ C | ☐ D |
| 5 | ☐ A | ☐ B | ☒ C | ☐ D |
| 6 | ☐ A | ☒ B | ☐ C | ☐ D |
| 7 | ☐ A | ☒ B | ☐ C | ☐ D |
| 8 | ☐ A | ☐ B | ☐ C | ☒ D |
| 9 | ☐ A | ☐ B | ☒ C | ☐ D |
| 10 | ☐ A | ☐ B | ☐ C | ☒ D |

SECTION II

Question 1:

(a) $x(x^3 - 27)$ (A1)

$= x(x-3)(x^2 + 3x + 9)$ (A1)

(b) $3x - 4 < 8$

$3x < 12$

$x < 4$ (A1)

OR

$-3x + 4 < 8$

$-3x < 4$

$x > -\frac{4}{3}$ (A1)

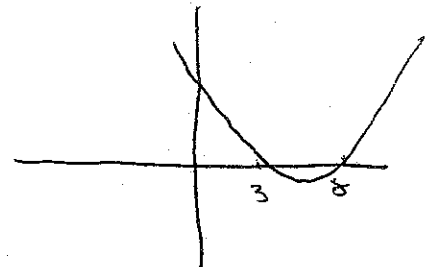
(c) $(x - 8)(x - 3) > 0$

1 mark / 2 marks
for JUST factorising

(A1)

(A1)

$x > 8$ or $x < 3$



(d) $\frac{4}{(m-2)(m+2)} - \frac{1}{m-2}$

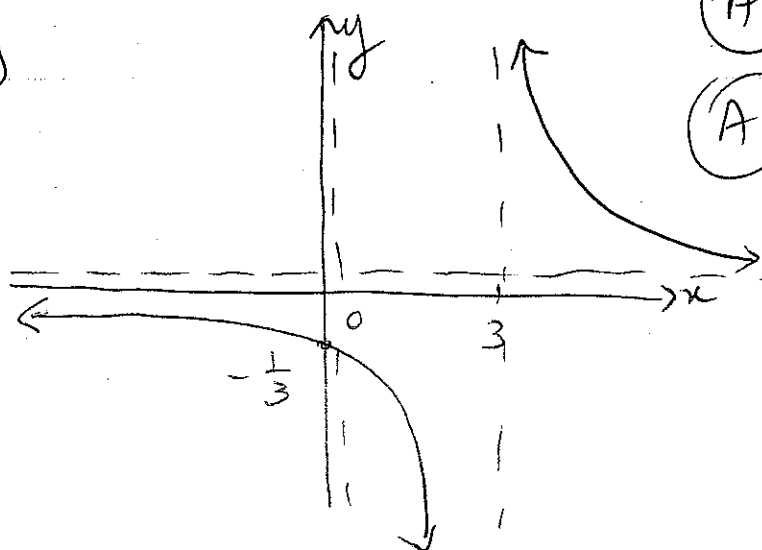
(A1)

$= \frac{4 - (m+2)}{(m-2)(m+2)}$ (A1)

$= \frac{2 - m}{(m-2)(m+2)}$ (A1)

Question 12:

(a) (i)



(A1)

shape

(A1)

Asymptotes.

(ii)

$$D = \{x \mid x \neq 3, x \in \mathbb{R}\}$$

(A1)

(b)

$$f\left(\frac{1}{a}\right) = \frac{1}{\left(\frac{1}{a}\right)^2} + \frac{2}{\frac{1}{a}} \quad (M1)$$

$$f\left(\frac{1}{a}\right) = a^2 + 2a \quad (A1)$$

(c) (i)

$$m = \frac{4-2}{2-5} \quad (M1)$$

$$m = \frac{2}{-3} \quad (A1)$$

$$(ii) \quad y + 2 = -\frac{2}{3}(x + 1) \quad (M1)$$

$$3y + 6 = -2x - 2$$

$$2x + 3y + 8 = 0$$

(M1)

$$(e) \quad \begin{aligned} 6a - 8b &= 22 \quad \text{--- (1)} \\ 6a + 9b &= -12 \quad \text{--- (2)} \end{aligned}$$

$$\textcircled{1} - \textcircled{2}$$

$$-17b = 34$$

$$b = -2$$

$$3a - 4(-2) = 11$$

$$3a = 3$$

$$a = 1$$

$$\left\{ \begin{aligned} a &= 1 \\ b &= -2 \end{aligned} \right\} \begin{aligned} &\textcircled{A1} \\ &\textcircled{A1} \end{aligned}$$

(f)

$$\frac{\sqrt{5}(\sqrt{5}+2)}{(\sqrt{5}-2)(\sqrt{5}+2)} \quad \textcircled{M1}$$

$$= \frac{5 + 2\sqrt{5}}{5 - 4}$$

$$5 + 2\sqrt{5} = a + \sqrt{b}$$

$$a = 5 \quad \textcircled{A1}$$

$$b = 20 \quad \textcircled{A1}$$

$$(iii) \quad d = \sqrt{(2-0)^2 + (4-0)^2}$$

$$d = \sqrt{4 + 16} \quad (M1)$$

$$d = \sqrt{20}$$

$$d = 2\sqrt{5} \text{ units} \quad (A1)$$

$$(iv) \quad x^2 + y^2 = (2\sqrt{5})^2$$

$$x^2 + y^2 = 20. \quad (A1)$$

$$(v) \quad (0,0) = \left(\frac{x+2}{2}, \frac{y+4}{2} \right) \quad (M1)$$

$$\frac{x+2}{2} = 0 \quad \left| \quad \frac{y+4}{2} = 0 \right.$$

$$x = -2 \quad \left| \quad y = -4 \right.$$

$$C(-2, -4) \quad (A1)$$

$$(vi) \quad d = \frac{|5(2) + 3(2) + 8|}{\sqrt{2^2 + 3^2}} \quad B(5, 2)$$

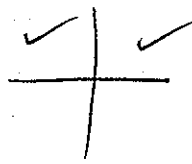
$$d = \frac{|10 + 6 + 8|}{\sqrt{13}}$$

$$\frac{24}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}} = \frac{24\sqrt{13}}{13}$$

$$d = \frac{24}{\sqrt{13}} \text{ units} \quad (A1)$$

Question 13

(a) (i) $\sin 30^\circ = \frac{1}{2}$

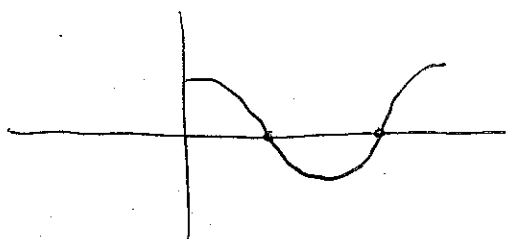


$\theta = 30^\circ, 150^\circ$

(A1)

(A1)

(ii) $2\theta = 90^\circ, 270^\circ, 450^\circ, 630^\circ$ $0^\circ \leq 2\theta \leq 720^\circ$



$\therefore \theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

(A1)

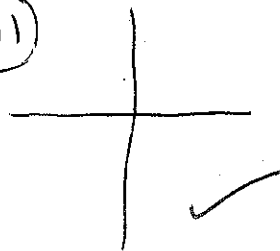
(A1)

(iii) $\tan 60^\circ = \sqrt{3}$

(M1)

$\therefore \theta = 300^\circ$

(A1)



(b)

$$\frac{\sin^2 \theta (1 + \cos \theta) - \sin^2 \theta (1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)}$$

$$= \frac{2 \sin^2 \theta \cos \theta}{1 - \cos^2 \theta}$$

(M1)

$$= \frac{2 \sin^2 \theta \cos \theta}{\sin^2 \theta}$$

$$= 2 \cos \theta$$

(A1)

(c) (i) If $f(x)$ is odd, $f(-x) = -f(x)$

$$f(-x) = -x \sqrt{4 - (-x)^2} \quad (M1)$$

$$= -x \sqrt{4 - x^2}$$

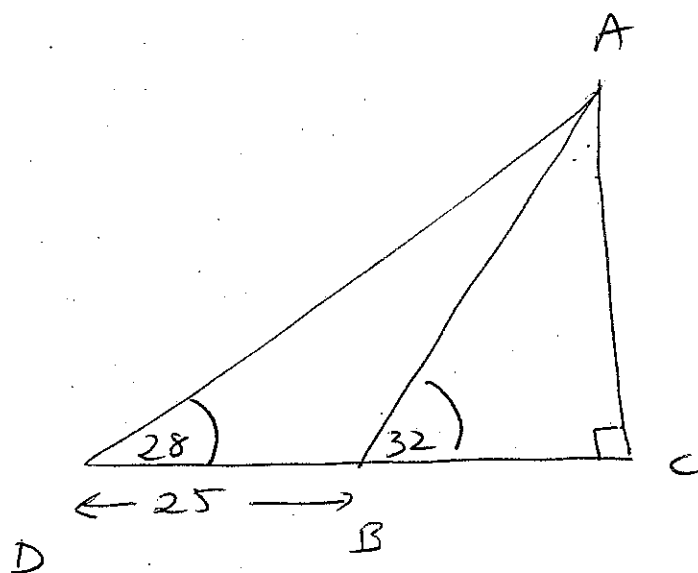
$$= - (x \sqrt{4 - x^2}) \quad (M1)$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function.

(ii) $\therefore f(1-x) = -\sqrt{3} \quad (A1)$

(d)



Since $\angle ABC = 32^\circ$

$$\therefore \angle DBA = 180^\circ - 32^\circ$$

$$= 148^\circ$$

* a variation is acceptable

(A1)

$\therefore$ $\angle$ sum of $\triangle DAB$

$$28^\circ + 148^\circ + \angle DAB = 180^\circ$$

$$\angle DAB = 180^\circ - 176^\circ = 4^\circ$$

$$(ii) \frac{BA}{\sin 28^\circ} = \frac{25}{\sin 40^\circ} \quad (M1)$$

$$BA = \frac{25 \sin 28^\circ}{\sin 40^\circ}$$

$$BA = 18.26$$

$$BA = 18.3 \text{ m} \quad (A1)$$

$$(iii) \frac{h}{18.26} = \sin 32^\circ$$

$$h = 18.26 \times \sin 32^\circ$$

$$h = 9.68$$

$$h = 9.7 \text{ m} \quad (A1)$$

Question 14:

$$(a) \quad \begin{aligned} \text{RHS} &= Ax^2 - Ax + Bx - B + C \\ &= Ax^2 + (B+A)x + (C-B) \end{aligned}$$

$$\boxed{2 = A} \quad (A1)$$

$$B + A = 1$$

$$\boxed{B = -1} \quad (A1)$$

$$1 = C - B$$

$$\boxed{0 = C} \quad (A1)$$

$$(b) \quad (i) \quad \alpha + \beta = \frac{7}{2} \quad (A1)$$

$$(ii) \quad \begin{aligned} \alpha\beta &= \frac{12}{2} \\ &= 6 \end{aligned} \quad (A1)$$

$$(iii) \quad \frac{\beta + \alpha}{\alpha\beta} \quad (M1) = \frac{\frac{7}{2}}{6} = \frac{7}{12} \quad (A1)$$

$$(iv) \quad \begin{aligned} \alpha\beta(\alpha + \beta) &= 6\left(\frac{7}{2}\right) \\ (M1) \quad &= 21 \quad (A1) \end{aligned}$$

$$\begin{aligned} (v) \quad &(\alpha + \beta)^2 - 2\alpha\beta \quad (M1) \\ &= \left(\frac{7}{2}\right)^2 - 2(6) \\ &= \frac{49}{4} - 12 \quad (A1) \\ &= \frac{5}{4} \end{aligned}$$

$$(c) (i) \quad \Delta = [-2(k+1)]^2 - 4(4k+1)(1-2k) \quad (M1)$$

$$\Delta = 4(k^2 + 2k + 1) - 4(4k + 1 - 8k^2 - 2k)$$

$$\Delta = 4k^2 + 8k + 4 - 16k - 4 + 32k^2 + 8k$$

$$\Delta = 36k^2 \quad (A1)$$

$$(ii) \quad \alpha + \beta = \frac{-(k+1)}{(k-1)}$$

$$\alpha\beta = \frac{-2}{k-1}$$

$$\alpha + \frac{1}{\alpha} = \frac{-(k+1)}{k-1}$$

$$\alpha \times \frac{1}{\alpha} = \frac{-2}{k-1} \quad (M1)$$

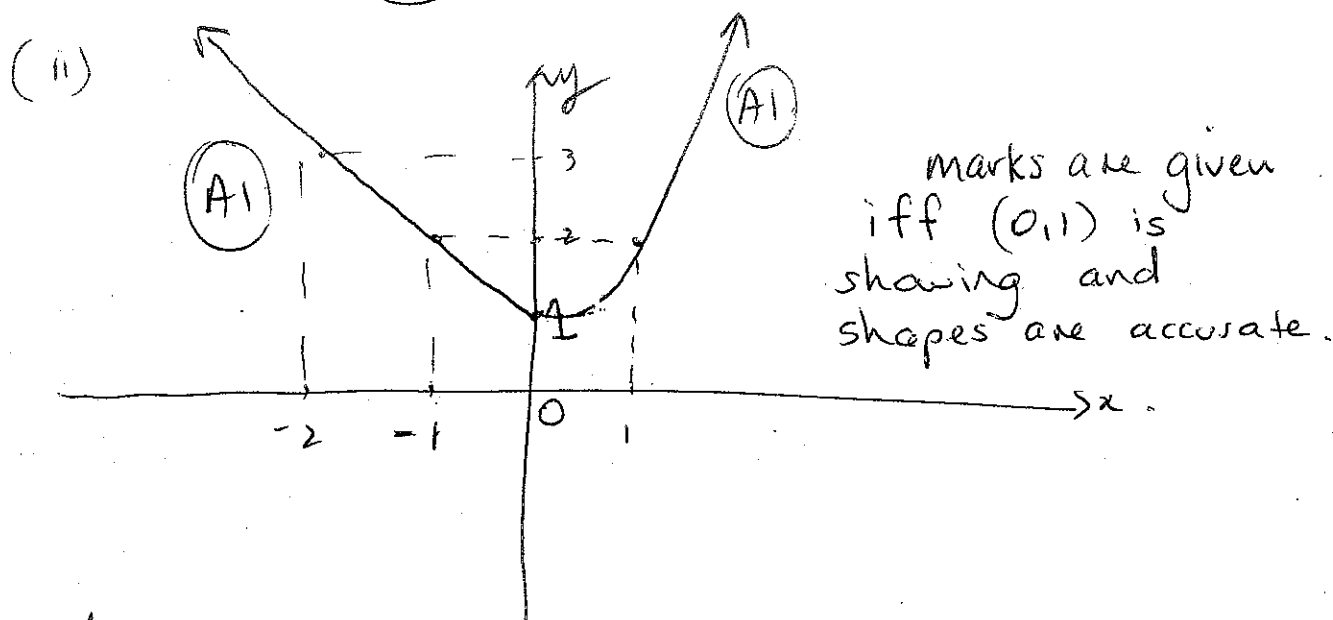
$$1 = \frac{-2}{k-1}$$

$$\boxed{k-1 = -2}$$

$$\boxed{k = -1} \quad (A1)$$

Question 15:

(a) (i) $[1 - (-2)] + [9 + 1]$ (M1)
 $= 3 + 10$
 $= 13$ (A1)



(b) $\frac{dy}{dx} = 20x^3 - 3x^{-2} + x^{-1/2}$

$\frac{dy}{dx} = 20x^3 - \frac{3}{x^2} + \frac{1}{\sqrt{x}}$
(A1) (A1) (A1)

(c) $\frac{dy}{dx} = \frac{(3-x) \cdot (2x) - (x^2+1)(-1)}{(3-x)^2}$ (M1)
 $= \frac{6x - 2x^2 + x^2 + 1}{(3-x)^2}$
 $= \frac{6x + 1 - x^2}{(3-x)^2}$ (A1)

$$(d) \quad f'(x) = 4(2x-5)^3 \times 2$$

$$f'(x) = 8(2x-5)^3 \quad (M1)$$

$$\therefore 0 = 8(2x-5)^3$$

$$0 = 2x-5$$

$$5 = 2x$$

$$x = 5/2 \quad (A1)$$

$$(e) \quad \frac{dy}{dx} = 3x^2 + 5x$$

$$m = 8 \quad (A1)$$

$$x = 1$$

$$y = 2$$

$$y - 2 = 8(x - 1)$$

$$y - 2 = 8x - 8$$

$$y = 8x - 6 \quad (A1)$$

$$(f) \quad \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3} \quad (M1)$$

$$= \lim_{x \rightarrow 3} x+3$$

$$= 6 \quad (A1)$$

Question 16:

(a) $\angle EFB = 180^\circ - 141^\circ$ — co-interior $\angle$'s in 11 lines
 $\angle EFB = 39^\circ$ (M1) $AB \parallel EF$ (are supplementary)

$\angle EFC = \angle EFB + \angle BFC = 68^\circ$ — alt $\angle$'s in 11 lines
(M1) $CD \parallel EF$ are (=)

$68^\circ = 39^\circ + y^\circ$
 $29^\circ = y^\circ$ (A1)

Aim: To Prove $\triangle RST \equiv \triangle QRP$.

(b)(i) Proof: $\angle STR = \angle PRS$ — alt $\angle$'s in 11 lines
(M2) $\angle SRT = \angle PQR$ — alt $\angle$'s in 11 lines $ST \parallel RP$ are (=),
 $RS \parallel PQ$ are (=)

$\therefore \triangle RST \equiv \triangle QRP$

(equiangular) (A1)

(ii) $\frac{ST}{8} \times \frac{3}{12}$ | $\frac{ST}{RP} = \frac{RS}{PQ}$
12 ST = 24 (M1)

ST = 2 cm. (A1)

(c)

$$\sin 150^\circ \sec (-150^\circ)$$

$$\left(\frac{1}{2}\right) \times \frac{1}{-\sqrt{3}/2} \quad (M1)$$

$$= \frac{1}{2} \times -\frac{2}{\sqrt{3}}$$

$$= -\frac{1}{\sqrt{3}} \quad (A1)$$

(d)

$$x^2 - 6x + 9 = -12y$$

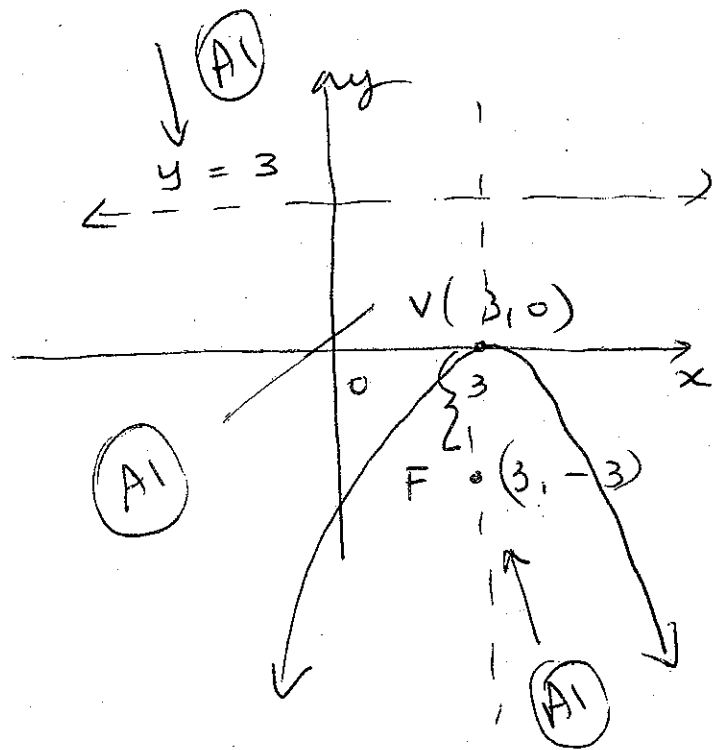
$$(x-3)^2 = -12y$$

$$V(3,0)$$

$$a = 3$$

$$F(3,-3)$$

$$y = 3$$



(e)

$$P(x,y)$$

$$A(2,y)$$

$$PA = PB$$

$$\sqrt{(x-2)^2 + (y-y)^2} = \sqrt{(x-2)^2 + (y+4)^2}$$

$$\sqrt{(x-2)^2} = \sqrt{(y+4)^2}$$

$$\pm(x-2) = y+4$$

$$(A1) \quad y = x - 6 \quad \text{OR}$$

$$(A1) \quad y = -x - 2$$

$$B(x,-4)$$

